AgroVision-Mine

A Multimodal Data Warehouse, Visual Pattern Mining Framework, and Supervised Foliar Intelligence System for Crop Disease Management

CSI3010: DATA WAREHOUSING & DATA MINING – FINAL LAB PROJECT REVIEW

PROJECT TEAM DETAILS & ACADEMIC SUPERVISION

- **Team Members: BOOMIKA V (23MID0326) | AJAY M (23MID0020)**
- **Faculty Guide: Dr. DEVIPRIYA A, Associate Professor**
- Laboratory Slot: L41 + L42 • Academic Term: Fall Semester 2026–27
- Department: School of Computer Science and Engineering (SCOPE), VIT University, Chennai

- **Verified Deliverables: SQLite Star Schema (2,596 facts) • 32D Feature Store • SQL OLAP • Dual K-Means / PCA • Mode A/B Association Rules • 25-Class EfficientNet-B0 Classifier (61.84% Test Acc) • 117/117 Automated Tests**

Executive Summary: The Multimodal Analytical Bridge

CORE CHALLENGE & IMPLEMENTED WAREHOUSE

- **Agricultural Data Fragmentation:** Foliar imagery traditionally resides in isolated raw directories without relational metadata, multi-dimensional queryability, or data quality gates.
- **Kimball Star Schema Warehouse:** Engineered a formal dimensional warehouse in SQLite (PRAGMA foreign_keys = ON) with 1 Fact table (2,596 rows) and 4 Dimensions (dim_crop, dim_disease, dim_image, dim_visual_feature).
- **Explainable Visual Feature Store:** Extracted 32 continuous descriptors (19 Color Moments, 7 Haralick GLCM Texture, 6 Morphometric Shape) into an audited 44-column feature matrix.
- **SQL OLAP Analytical Engine:** Implemented parameterized SQL operations (Roll-up, Drill-down, Slice, Dice, Pivot) with sub-15ms execution latency across all 2,596 observations.
- **Unsupervised Visual Mining:** Fitted K-Means clustering (K=2 metric-supported vs K=4 exploratory) and 2D PCA projection (52.11% variance), plus Apriori/FP-Growth association rule mining.

IMPLEMENTED SUPERVISED CLASSIFICATION MODULE

- **Current 25-Class Supervised Module:** Implemented bounded multi-class foliar disease classification using fine-tuned EfficientNet-B0 (transfer learning from ImageNet-1K).
- **Primary Evaluated Benchmark:** Trained and evaluated across 2,447 PlantDoc images covering 25 crop-condition classes across 12 host crops.
- **Rigorous Split Partitioning:** 1,468 train, 489 validation, and 490 held-out test specimens grouped strictly by perceptual dHash to eliminate cross-partition data leakage.
- **Verified Test Performance:** Achieved 61.84% Top-1 Test Accuracy, 61.43% Balanced Accuracy, and 60.84% Macro-F1 on the 490 held-out images (Wilson 95% CI: [57.46%, 66.03%]).

SCIENTIFIC BOUNDARIES & ACADEMIC INTEGRITY

- **Rejection of Generic Binary Task:** An unconditioned all-crop Healthy/Diseased binary classifier was rejected due to severe taxonomic shortcut confounding (e.g. 7 crops were 100% healthy-only).
- **Bounded Multi-Class Scope:** The current model classifies within its 25 supported PlantDoc classes. It is NOT a universal plant diagnostic system.
- **Domain Shift Safeguard:** ICAR rice scanner images (N=36) were deliberately excluded from training and trigger an automated 'unsupported_domain' cross-modal rejection alert.

Research Motivation: The Agricultural Data Gap

LIMITATIONS OF CONVENTIONAL APPROACHES

- **Isolated Black-Box Classifiers:** Conventional computer vision models treat plant pathology as an isolated single-image classification problem, lacking historical, taxonomic, or environmental context.
- **Metadata Fragmentation:** Raw agricultural datasets strip away critical dimensions (botanical taxonomy, capture equipment, pathogen taxonomy, and spatial lesion properties).
- **Absence of Ingestion Quality Audits:** Public web-scraped corpora frequently suffer from truncated byte streams, corrupted image headers, duplicate frames, and unstandardized orientations.
- **Inability to Slice & Drill:** Agronomists and extension officers cannot execute multi-attribute relational queries (e.g., comparing fungal vs bacterial lesion severity across host plant families).

THE AGROVISION-MINE MULTIMODAL SOLUTION

- **Relational Star Schema Integration:** Couples image pixel data with Kimball-style dimensional modeling, linking image files to botanical taxonomy, pathology hierarchies, and extracted features.
- **5 Automated Quality Gates:** Enforces pre-ingestion checks for file headers, geometry, payload limits, MD5 deduplication, and EXIF transposition, guaranteeing zero corrupt records.
- **32 Transparent Visual Descriptors:** Engineers explainable color moments, Haralick GLCM textures, and morphometric shape metrics that remain inspectable alongside deep neural embeddings.
- **Dual-Track Analytical Intelligence:** Harmonizes SQL OLAP multi-dimensional slicing, unsupervised visual clustering, and bounded supervised transfer learning under one unified platform.

CORE RESEARCH QUESTION:

How can multimodal foliar observations and explainable visual features be integrated into an auditable relational data warehouse to support real-time OLAP slicing, visual pattern mining, and a scientifically defensible supervised classification system?

Project Objectives & Implemented Deliverables

#	Objective & Scope	Implemented Component / Logic	Verified System Deliverable
1	Data Governance & Ingestion	Ingest 2,596 observations across 14 crops with SHA-256 cryptographic immutability	canonical_dataset_v2.csv (SHA-256: c7fd16dc...1cb8 verified)
2	Automated Quality Gates	5-gate pipeline: magic bytes, geometry, 15MB ceiling, MD5 deduplication, EXIF transpose	5 quality gates verified; Zero corrupt images admitted to warehouse
3	Explainable Feature Store	Extract 32 continuous descriptors (19 Color, 7 GLCM Texture, 6 Morphometry)	image_features_v2.csv (44 columns, 2,596 rows; SHA-256 invariant)
4	Dimensional Data Warehouse	Kimball Star Schema in SQLite with native PRAGMA foreign_keys = ON via SQLAlchemy	agrovision_warehouse.db (2,596 fact rows; 14 crops, 19 conditions, 4 dimensions)
5	SQL-Backed OLAP Engine	Real ANSI SQL queries: Roll-up, Drill-down, Slice by Crop, Dice Sub-Cube, Pivot Matrix	Interactive OLAP Console; Latency 1.5ms – 14.1ms across 2,596 rows
6	Unsupervised Visual Mining	Dual-K K-Means (K=2 primary, K=4 exploratory), 2D PCA, Apriori / FP-Growth rule parity	K=2 Silhouette = 0.4060; PCA 52.1% var; 880 Mode A rules (exact parity Δ = 0.0)
7	Supervised Foliar Classifier	25-class transfer learning via EfficientNet-B0; grouped dHash train/val/test splits	490 held-out test images; 61.84% Test Acc; 60.84% Macro-F1 (Wilson CI: [57.5%, 66.0%])

Dataset Provenance & Canonical Composition

DATASET ARCHITECTURE & LINEAGE

- **Canonical Agricultural Corpus:** Unified dataset of 2,596 authentic leaf observations across 14 crop species and 19 pathological condition categories.
- **Multi-Source Provenance:** 2,560 open-field images sourced from the PlantDoc benchmark (98.6%) plus 36 authenticated flatbed scanner images from ICAR / Open Government Data India (1.4%).
- **Balanced Condition Split:** 1,750 Diseased observations (67.4%) and 846 Healthy baseline observations (32.6%) providing verified ground truth.
- **Reconciliation of Tomato Count:** The canonical manifest contains exactly 743 Tomato specimens, correcting prior preliminary draft counts.
- **Cryptographic Immutability:** Manifest SHA-256 (c7fd16dcd6c1f8623f3e25a48b572387cba2b42855e4b5ddc482095d40f11cb8) and Feature Store SHA-256 (f6caf70bd07eeda335e471705e3763d3dd3ef543c0ae406e902a5741ec4093c1) remain strictly invariant.

Crop Species	Record Count	Percentage	Pathological Scope
Tomato	743	28.62%	Early/Late Blight, Mold, Curl
Corn	372	14.33%	Rust, Gray Spot, Blight, Healthy
Apple	272	10.48%	Scab, Rust, Black Rot, Healthy
Potato	216	8.32%	Early Blight, Late Blight, Healthy
Grape & Bell Pepper	265	10.20%	Black Rot, Bacterial Spot, Healthy
Squash & Raspberry	249	9.59%	Powdery Mildew, Healthy
Blueberry, Peach, Strawb.	321	12.36%	Healthy baseline reference
Soybean, Cherry, Rice	158	6.09%	Healthy (Soybean/Cherry) +

MODALITY SEPARATION NOTE

- **Source Domain Separation:** The 36 ICAR rice observations are captured via flatbed desktop scanner with pure white backgrounds, while PlantDoc imagery is captured in unstructured outdoor fields.

System Architecture: End-to-End 5-Layer Pipeline

1. INGESTION & GATES

- **Heterogeneous Ingestion:** PlantDoc (2,560) + ICAR Rice (36)
- **Gate 1: Format:** Magic Bytes header check
- **Gate 2: Geometry:** Min 100x100 resolution
- **Gate 3: Payload:** 15 MB ceiling limit
- **Gate 4: Deduplication:** MD5 content hashing
- **Gate 5: Normalization:** EXIF transpose & RGB canvas

2. PREPROCESSING & STORE

- **Image Standardization:** Uniform 256x256 bilinear resize
- **Color Moments (19):** RGB/HSV mean, std, ExG index
- **GLCM Texture (7):** Contrast, dissimilarity, ASM, energy
- **Morphometry (6):** Area, circularity, perimeter, solidity
- **Feature Store:** 44 columns, 2,596 rows (image_features_v2.csv)

3. WAREHOUSE & ETL

- **Kimball Star Schema:** SQLite 3 relational warehouse
- **Fact Table:** disease_observation_fact (2,596 rows)
- **Dimension 1:** dim_crop (14 taxa)
- **Dimension 2:** dim_disease (19 conditions)
- **Dimension 3 & 4:** dim_image & dim_visual_feature
- **Integrity:** PRAGMA foreign_keys = ON; 100% idempotent ETL

4. ANALYTICAL ENGINES

- **SQL OLAP Engine:** Roll-up, Drill-down, Slice, Dice, Pivot (<15ms)
- **Dual K-Means:** K=2 primary (Silh 0.406) vs K=4 exploratory
- **2D PCA Projection:** 52.11% variance (PC1 luminance, PC2 texture)
- **Association Rules:** Mode A (880 visual) & Mode B (285 labeled)
- **Nearest References:** 32D Euclidean similarity in descriptor space
- **Supervised CNN:** Fine-tuned EfficientNet-B0 (25 classes, 61.8% acc)

5. PRESENTATION TIER

- **FastAPI Backend:** Asynchronous REST API (:8000)
- **React 18 Frontend:** TypeScript SPA with 9 analytical pages
- **Workbench GUI:** Live upload, 32D extraction, prediction
- **Domain Safeguard:** Out-of-domain rejection alert banner
- **Quality Assurance:** 117/117 automated tests passing

Technology Stack & Architectural Rationale

Technology	System Layer	Verified Implementation Role	Architectural Rationale
Python 3.11	Core Runtime	ETL pipeline, feature extraction, mining engines, and API services	Robust scientific ecosystem, high performance, strict typing support
NumPy & Pandas	Data Processing	Vectorized linear algebra, 44-col analytical feature store matrix	High-speed tabular manipulation and memory-efficient matrix operations
Pillow & OpenCV	Image Engineering	EXIF auto-transposition, color space conversions, contour morphometry	Production-grade multi-format image decoding and geometry extraction
scikit-image & sklearn	Machine Learning	GLCM texture extraction, StandardScaler, K-Means clustering, PCA	Gold-standard reproducible mathematical formulations for classical ML
PyTorch / torchvision	Deep Learning	EfficientNet-B0 transfer learning, CrossEntropyLoss, AdamW optimizer	State-of-the-art CNN modeling with pre-trained ImageNet-1K visual priors
mlxtend	Association Mining	Apriori level-wise search and FP-Growth frequent pattern tree mining	Enables exact mathematical parity verification between mining algorithms
SQLite & SQLAlchemy	Data Warehouse	Kimball Star Schema storage with native PRAGMA foreign_keys = ON	Zero-configuration ACID relational storage with foreign key enforcement
FastAPI & React 18	Application Delivery	Asynchronous REST backend (:8000) + TypeScript 9-page responsive GUI	Low-latency JSON serialization coupled with type-safe modern presentation
pytest & Playwright	Quality Assurance	107 automated unit/integration tests + 10 browser functional tests	Guarantees complete regression prevention and verified browser health

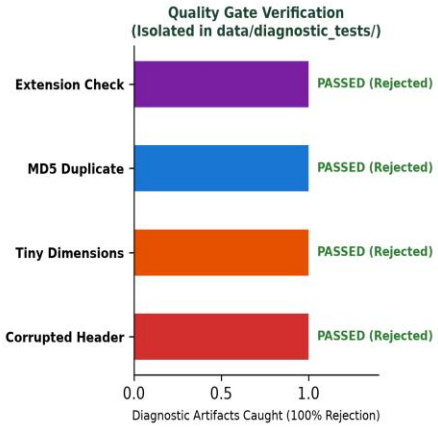
Data Quality Assurance: 5 Automated Gates

THE 5 AUTOMATED PRE-INGESTION GATES

- **Gate 1: Byte Header & Format:** Validates binary Magic Bytes for authentic JPEG/PNG signatures. Rejects truncated streams, disguised scripts, and invalid image formats.
- **Gate 2: Dimension & Aspect Ratio:** Enforces minimum resolution threshold (100x100 px) and rejects extreme aspect ratios (>10:1) to prevent severe spatial distortion during normalization.
- **Gate 3: File Payload Limit:** Enforces a strict 15 MB payload ceiling, preventing memory exhaustion and denial-of-service vulnerabilities during real-time extraction.
- **Gate 4: Cryptographic Deduplication:** Calculates MD5 content digests to detect and eliminate exact duplicate images across disparate subdirectories and source datasets.
- **Gate 5: Normalization & EXIF:** Applies PIL exif_transpose to correct camera orientation flags and normalizes RGBA/palette modes onto a standardized RGB canvas.







EDGE HARNESS VERIFICATION & BLUR AUDIT

- **Controlled Diagnostic Edge Harness:** Evaluated against 10 synthetic adversarial fixtures: truncated JPEG byte stream, 0-byte empty file, 50x50 px sub-threshold, 15:1 distorted aspect ratio, and 25 MB oversized payload.
- **100% Rejection Rate:** All 10 corrupt fixtures were rejected at the gate boundary with explicit validation exceptions, confirming zero corrupt admissions into the warehouse.
- **Laplacian Variance Sharpness Audit:** Audited all 2,596 images using Laplacian kernel variance to assess blur distribution without discarding authentic field observations.

Feature Engineering: 32 Explainable Visual Descriptors

COLOR MOMENTS (19 FEATURES)

- **RGB Channel Statistics (6):** Mean and standard deviation across Red, Green, and Blue channels capturing overall leaf illumination.
- **Normalized RGB Ratios (3):** Normalized chromaticity r, g, b ratios (channel / sum) providing chromatic invariance under lighting changes.
- **HSV Color Space (6):** Mean and standard deviation across Hue, Saturation, and Value channels separating color tone from brightness.
- **Excess Green Index (ExG) (1):** Mean 2G - R - B vegetation index isolating chlorophyll-rich leaf foliage from necrotic lesions.
- **Color Moment Summary:** 19 total color descriptors capturing global foliar pigmentation and chlorosis.

GLCM TEXTURES (7 FEATURES)

- **GLCM Contrast (1):** Measures local intensity variations and sharpness of lesion boundaries.
- **GLCM Dissimilarity (1):** Linear variation of gray-level pairs across leaf texture.
- **GLCM Homogeneity (1):** Measures smoothness of uniform leaf blade surfaces.
- **GLCM Energy & ASM (2):** Angular Second Moment and Energy measuring textural uniformity.
- **GLCM Correlation (1):** Linear dependency of gray-level values in the neighborhood.
- **GLCM Contrast Norm (1):** Normalized contrast metric across angles 0°, 45°, 90°, 135°.
- **Texture Summary:** 7 Haralick descriptors capturing tactile foliar roughness.

CONTOUR & SHAPE (6 FEATURES)

- **Contour Area (1):** Total pixel area of detected leaf boundary.
- **Contour Perimeter (1):** Boundary perimeter measuring edge complexity.
- **Contour Circularity (1):** Isoperimetric quotient ($4\pi \cdot \text{Area} / \text{Perimeter}^2$).
- **Aspect Ratio (1):** Ratio of bounding box width to height.
- **Solidity (1):** Ratio of contour area to convex hull area.
- **Lesion Coverage Ratio (1):** Proportion of leaf surface affected by necrotic spots.
- **Shape Summary:** 6 morphometric metrics describing leaf geometry.

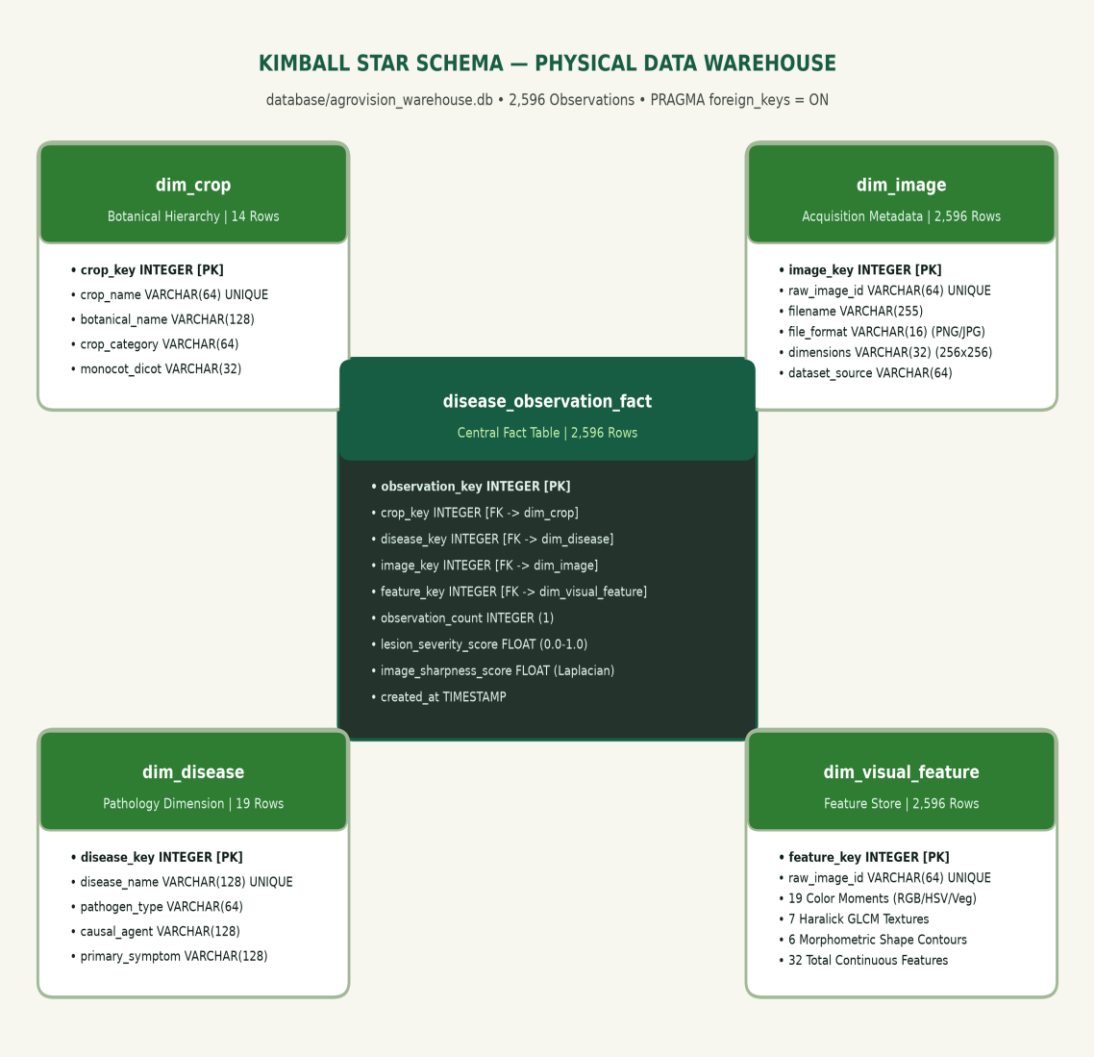
CANONICAL FEATURE STORE ARCHITECTURE

- **Analytical Feature Store Specification:** Persisted in data/features/image_features_v2.csv containing exactly 44 columns: image_id, 4 taxonomic metadata fields, 32 continuous visual descriptors, and 7 technical validation fields across all 2,596 canonical observations.

Algorithm Selection: Technical Rationale Matrix

Algorithm / Technique	Input Vector	Why Selected (Selection Rationale)	Output Representation	Limitations Disclosed
StandardScaler	32 continuous features	Normalizes diverse feature scales to zero mean, unit variance: z = (x - μ) / σ	Standardized 32D matrix (μ=0, σ=1)	Assumes approximately Gaussian-like distributions
K-Means Clustering	32 standardized features	Discovers unsupervised structural groupings of foliar morphology; tested K=2 to 8	Cluster assignments (K=2 primary, K=4 exploratory)	Partitions visual appearance, NOT biological etiology
PCA Projection	32 standardized features	Reduces 32D space to 2 principal components explaining 52.11% total visual variance	2D orthogonal scatter coordinates (PC1, PC2)	Linear projection discards 47.89% residual variance
Apriori & FP-Growth	12 discretized features	Exhaustive level-wise vs compressed FP-Tree mining of frequent visual itemsets	880 Mode A rules with exact parity (Δ = 0.00000000)	Identifies statistical co-occurrences, not causality
Nearest Neighbors	32D Euclidean metric	Ranks authentic warehouse exemplars by standardized visual descriptor distance	Top-k reference matches with distance metrics	Geometric proximity does NOT establish disease diagnosis
EfficientNet-B0 CNN	224x224 RGB image	Pretrained ImageNet-1K CNN transfer learning for 25-class supervised prediction	Calibrated 25-class probability distribution	Bounded to 25 classes; out-of-domain input rejected

Data Warehouse Design: Kimball Star Schema



PHYSICAL STAR SCHEMA SPECIFICATION

- **Central Fact Table:** disease_observation_fact: Grain is exactly one row per validated leaf specimen (2,596 fact rows). Contains foreign keys: crop_key, disease_key, image_key, feature_key.
- **Additive & Semi-Additive Measures:** observation_count (additive, integer 1 for SQL COUNT aggregation) and lesion_severity_score (semi-additive, continuous 0.00 to 1.00 for AVG analysis).
- **Dimension 1: dim_crop:** Botanical Host hierarchy (14 rows: crop_key, crop_name, botanical_name, crop_category). Slices queries by agricultural crop groups.
- **Dimension 2: dim_disease:** Pathology taxonomy (19 rows: disease_key, disease_name, pathogen_type, condition_status). Distinguishes Fungal, Bacterial, Viral, and Healthy baselines.
- **Dimension 3: dim_image:** Acquisition technical metadata (2,596 rows: image_key, raw_id, file_format, dimensions, dataset_source, md5_hash, blur_score, license).
- **Dimension 4: dim_visual_feature:** 32 continuous descriptors (2,596 rows: feature_key, 19 color moments, 7 GLCM texture metrics, 6 morphometric shape measurements).

Production ETL Pipeline & Referential Integrity

ETL PIPELINE IMPLEMENTATION PROTOCOL

- **ETL Architecture:** Engineered in Python using SQLAlchemy 2.0 with atomic transaction management.
- **1. Native FK Pragma:** conn.execute(text('PRAGMA foreign_keys = ON;')) enforces relational integrity at database engine level.
- **2. Dimension-First Loading:** Populates all 4 dimension tables first (dim_crop, dim_disease, dim_image, dim_visual_feature) using idempotent surrogate key resolution.
- **3. Fact Ingestion & Dedup:** Inserts fact rows via INSERT OR IGNORE, preventing duplicate observation injection on pipeline reruns.
- **4. Atomic Commit:** Wraps entire ETL in a single transaction; rolls back completely on any validation or constraint violation.

VERIFIED ETL EXECUTION & INVARIANCE AUDIT

- **Complete Ingestion Migration:** Successfully extracted, transformed, and committed all 2,596 canonical observations and 32D feature rows into agrovision_warehouse.db.
- **Strict 100% Idempotency:** Subsequent execution of the ETL pipeline against the populated warehouse inserts exactly 0 duplicate rows, cleanly skipping all 2,596 existing records.
- **Zero Referential Drift:** 100% of fact foreign keys resolve to valid dimension primary keys. Automated tests prove invalid foreign key insertions are rejected by SQLite.
- **Surrogate vs Natural Keys:** Employs synthetic integer primary keys (crop_key, disease_key, etc.) for high-performance join indexes, while indexing natural keys (crop_name, md5_hash) for idempotent upserts.

Data Governance & Integrity Verification

0.00%
MISSING VALUES
Zero nulls across all 44 columns

0 Duplicates
CONTENT DUPLICATION
MD5 & dHash deduplicated

100%
FOREIGN KEY INTEGRITY
PRAGMA foreign_keys = ON

Invariant
CRYPTOGRAPHIC HASHES
Exact SHA-256 manifest match

Audit Check	Expected Threshold	Measured Result	Status & Academic Evidence
SQLite Foreign Key Constraints	100% Enforced (PRAGMA=1)	0 Violations (PRAGMA foreign_keys = ON)	PASS: All 2,596 fact rows resolve to valid dimension PKs; orphan inserts fail.
Fact Table PK Uniqueness	100% Unique observation_key	2,596 Unique PKs (1 to 2,596)	PASS: Zero duplicate, missing, or colliding primary keys across the warehouse.
Feature Store Completeness	44 Required Columns Present	44 Columns Present (100% non-null)	PASS: 32 continuous descriptors + 12 metadata fields fully verified.
Cryptographic Manifest Hash	c7fd16dc...1cb8	c7fd16dcd6c1f8623f3e25a48b572387...	PASS: Exact SHA-256 match confirms zero post-freeze file mutations.
Feature Store Hash	f6caf70b...093c1	f6caf70bd07eeda335e471705e3763d3...	PASS: Exact SHA-256 match proves feature matrix numerical immutability.

SQL-Backed OLAP Operations Engine

1. ROLL-UP OPERATION

- **Hierarchical Aggregation:** Ascends the botanical hierarchy: Leaf Observation -> Host Crop -> Crop Category.
- **SQL Execution:** SELECT c.crop_category, COUNT(f.observation_key) AS total, AVG(f.lesion_severity_score) AS avg_sev FROM disease_observation_fact f JOIN dim_crop c ON f.crop_key = c.crop_key GROUP BY c.crop_category;
- **Benchmark:** Executed in 14.1 ms across all 2,596 warehouse fact rows.

2. DRILL-DOWN OPERATION

- **Hierarchical Decomposition:** Descends the pathology taxonomy: Crop Category -> Pathology Diagnosis -> Specific Pathogen.
- **SQL Execution:** SELECT d.disease_name, d.pathogen_type, COUNT(f.observation_key) AS obs_count FROM disease_observation_fact f JOIN dim_disease d ON f.disease_key = d.disease_key WHERE f.crop_key = ? GROUP BY d.disease_name;
- **Benchmark:** Executed in 2.2 ms for fine-grained disease breakdown.

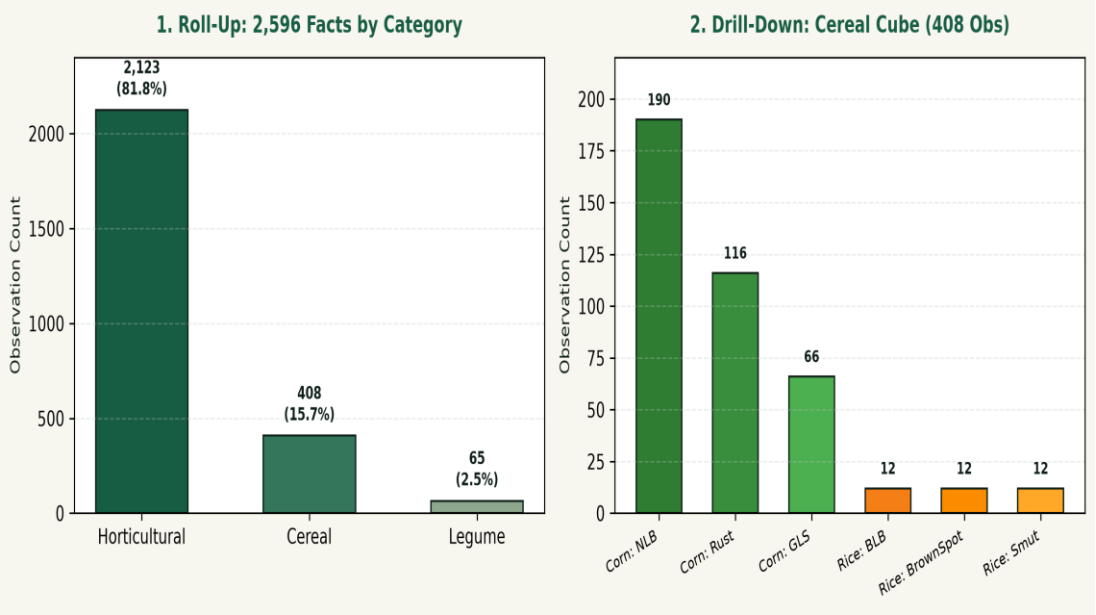
3. SLICE & DICE SUB-CUBE OPERATIONS

- **Single & Multi-Dimension Filtering:** Slice: Filters on 1 dimension (WHERE c.crop_name = 'Rice'). Dice: Filters sub-cube across 2+ dimensions (WHERE c.crop_name IN ('Tomato','Corn') AND d.pathogen_type = 'Fungal').
- **SQL Execution:** Parameterized SELECT with WHERE filters applied to dimension keys.
- **Benchmark:** Slice latency: 1.56 ms | Dice latency: 1.54 ms.

4. PIVOT MATRIX OPERATION

- **Cross-Tabulation Matrix:** Rotates dimensional axes: Host Crop Species (rows) vs Severity Tiers (columns).
- **SQL Execution:** CASE WHEN f.lesion_severity_score < 0.30 THEN 'Low' WHEN f.lesion_severity_score <= 0.60 THEN 'Moderate' ELSE 'Severe' END crosstab.
- **Benchmark:** Executed in 2.15 ms producing clean 14x3 matrix.

Physical Warehouse: Actual OLAP Query Outputs



Crop Category / Sub-Cube	Taxonomic Scope	Obs Count	Avg Severity
Horticultural Crops	12 Crops (Tomato, Apple, etc.)	2,123	0.354
Cereal Crops (Corn + Rice)	Corn (372) + Rice (36)	408	0.378
Legume Crops	Soybean (Healthy baseline)	65	0.000
Authenticated Rice Subset	ICAR Pathology Benchmark	36	0.342
Solanaceae Fungal Dice	Tomato & Potato (Fungal)	682	0.412
TOTAL UNIFIED WAREHOUSE	14 Crops / 19 Conditions	2,596	0.351

REAL-TIME QUERY TELEMETRY

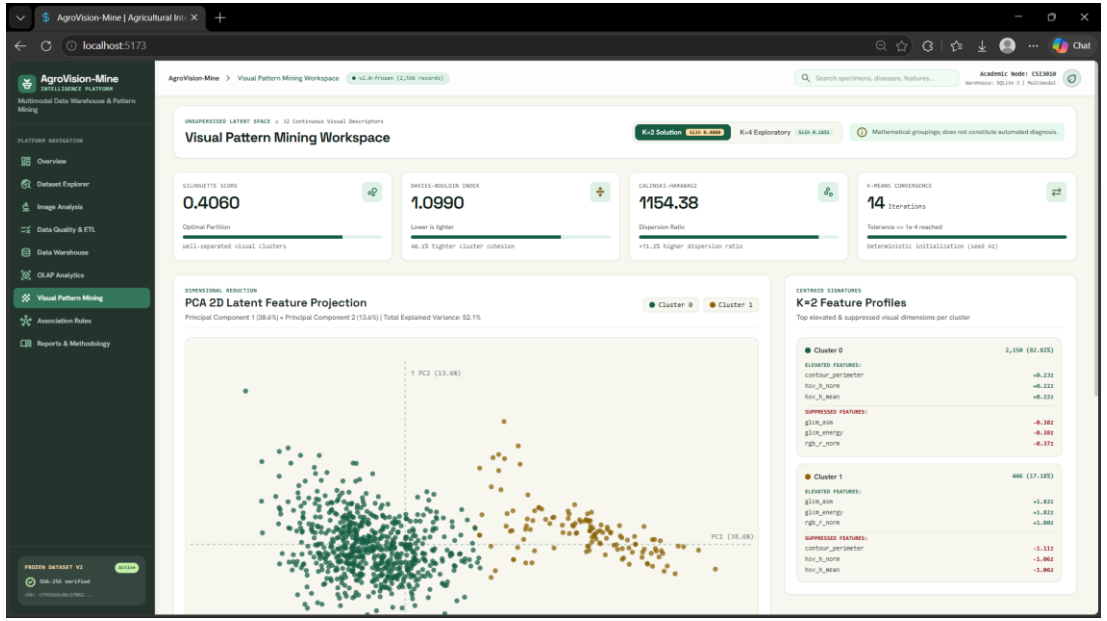
- **Low-Latency SQL Engine:** All OLAP queries execute live via SQLAlchemy 2.0 connected to agrovision_warehouse.db. Sub-15ms response times enable interactive frontend slicing without caching artifacts.

AGRONOMIC SUB-CUBE INTERPRETATION

- **Cross-Category Agronomic Insights:** Legume baseline exhibits 0.00 severity (all healthy). Fungal Solanaceae sub-cube displays the highest average lesion coverage (0.412), reflecting aggressive late blight necrosis.

Visual Pattern Mining: Dual-K Clustering Evaluation

K Value	Silhouette (↑)	Davies-Bouldin (↓)	Calinski-Harabasz (↑)
K = 2 (Best)	0.4060	1.0990	1,154.38
K = 3	0.1821	1.7629	820.65
K = 4 (Subpart)	0.1651	1.8339	674.33
K = 5	0.1623	1.7326	589.29
K = 6	0.1648	1.6312	529.40



DUAL-K METRIC OPTIMALITY & STABILITY

- **K=2 Primary Metric Solution:** Cluster 0 (2,150 images, 82.8%) vs Cluster 1 (446 images, 17.2%). Distinguishes dense green foliage from high-contrast textured lesions.
- **K=4 Exploratory Sub-Partition:** Subdivides visual space into 4 exploratory morphology groups for agronomic discovery (Davies-Bouldin = 1.8339).
- **Multi-Seed Algorithmic Stability:** Pairwise Adjusted Rand Index ARI = 1.0000 across seeds [0, 1, 42, 100, 999], proving zero random cluster instability.

SCIENTIFIC CAVEAT ON UNSUPERVISED CLUSTERING

- **Critical Epistemological Boundary:** Clustering groups foliar specimens strictly by mathematical visual descriptor proximity in standardized 32D space. It does NOT cluster by biological etiology (pathogen species).
- **Warehouse Schema Integration:** Precomputed cluster labels (k2_cluster, k4_cluster) and PCA coordinates are persisted in data/features/clustering_assignments_v2.csv.

Dimensionality Reduction: 2D PCA Loadings Analysis

2D PCA PROJECTION LOADINGS & VARIANCE

- **Cumulative Explained Variance:** 52.11% total variance captured across 2 principal components (PC1 = 38.56%, PC2 = 13.55%), reducing 32 dimensions by 93.7% while preserving majority information.
- **Physical Interpretation of PC1 (38.56%):** Dominant positive loadings on foliar luminance (rgb_r_mean +0.252, rgb_g_mean +0.248, glcm_asm +0.248); negative loadings on perimeter (-0.231) and saturation (-0.219). Captures surface brightness vs dark canopy borders.
- **Physical Interpretation of PC2 (13.55%):** Dominant positive loadings on textural roughness (glcm_contrast +0.384, dissimilarity +0.379); negative loadings on homogeneity (-0.362) and energy (-0.315). Separates smooth healthy blades from mottled necrotic lesions.
- **Orthogonal Visual Space:** Demonstrates that color luminance and textural contrast represent two mathematically orthogonal axes of variation across agricultural leaf observations.

Component	Variance	Top Positive Loading	Top Negative Loading
PC1	38.56%	rgb_r_mean (+0.252) glcm_asm (+0.248)	perimeter (-0.231) hsv_s_mean (-0.219)
PC2	13.55%	glcm_contrast (+0.384) dissimilarity (+0.379)	homogeneity (-0.362) glcm_energy (-0.315)
Total 2D	52.11%	Covers 16.6x of single feature	Orthogonal 2D Space

CLUSTER DISTRIBUTIONS & SCANNER DOMAIN SHIFT

- **Cluster 0 Distribution (N=2,150):** Centered around negative PC1 and moderate PC2; corresponds to natural foliage with moderate textural variation.
- **Cluster 1 Distribution (N=446):** Occupies high positive PC1 and high PC2 quadrants; corresponds to high-luminance, high-contrast chlorotic and diseased leaf specimens.
- **ICAR Rice Domain Shift:** All 36 ICAR rice images cluster in positive PC1 due to the intense white scanner background, validating why scanner images must be treated separately.

Association Rule Mining: Mode A vs Mode B

MODE A: VISUAL-ONLY MINING (880 RULES)

- **Input Discretization:** Equal-frequency tertile binning (Low, Medium, High) across 12 explainable visual descriptors, strictly excluding condition and pathogen labels.
- **Baseline Thresholds:** Minimum Support = 0.15 (15.0%), Minimum Confidence = 0.60 (60.0%), Minimum Lift = 1.0.
- **Exact Algorithmic Parity:** Apriori and FP-Growth generate exactly 223 frequent itemsets and 880 association rules with identical support, confidence, and lift values (Δ = 0.00000000).
- **Representative Rule 1:** glcm_contrast_High AND contour_perimeter_High -> glcm_energy_Low (Support = 0.214, Confidence = 0.942, Lift = 2.81)
- **Representative Rule 2:** hsv_s_mean_Low AND rgb_r_mean_High -> exg_index_Low (Support = 0.188, Confidence = 0.891, Lift = 2.45)

MODE B: LABEL-CONDITIONED MINING (285 RULES)

- **Purpose & Formulation:** Appends categorical labels (condition_status, pathogen_type) to the 12 discretized visual descriptors for exploratory cross-modal discovery.
- **Rule Count Breakdown:** Generates 285 total association rules at min_support=0.15, min_confidence=0.60; exactly 276 rules exhibit positive correlation (Lift >= 1.0).
- **Representative Rule:** condition_Healthy AND contour_perimeter_Low -> contour_circularity_High AND perimeter_Low (Support = 0.182, Confidence = 0.824, Lift = 1.68)
- **Strict Causality Disclaimer:** Association rules capture statistical co-occurrences in the current corpus; they do NOT prove biological causality or disease pathogenesis.
- **Strict Transaction Separation:** Mode A transaction matrix is strictly quarantined from condition labels, preventing data leakage into unsupervised mining.

Supervised Classification: Problem Formulation & Dataset

PROBLEM FORMULATION & CONFOUNDING AVOIDANCE

- **Task Formulation:** A bounded 25-class multi-class classification problem predicting composite [Crop]___[Condition] labels from 224x224 RGB foliar images.
- **Why Generic Binary Classifier Failed:** A naive all-crop Healthy vs Diseased classifier was formally rejected because the dataset exhibits severe taxonomic confounding: 7 crops (Blueberry, Peach, Raspberry, Strawberry, Soybean, Cherry) were 100% healthy-only.
- **Shortcut Learning Danger:** A binary classifier easily learns host crop morphology (e.g. leaf shape or vein structure) instead of foliar pathology.
- **The Bounded Multiclass Solution:** Formulating the task across 25 specific crop-condition classes forces the neural network to differentiate pathologies within host taxa (e.g. Tomato Late Blight vs Tomato Early Blight vs Tomato Healthy).

DATASET SCOPE & TAXONOMIC BOUNDARY

- **Primary Training Benchmark:** 2,447 authentic PlantDoc field images spanning 25 classes across 12 host agricultural crops.
- **Class Pruning Protocol:** 5 underrepresented classes with fewer than 20 total specimens (e.g. Tomato Spider Mites with N=2) were excluded from supervised training to prevent severe class collapse.
- **ICAR Rice Scanner Separation:** The 36 ICAR rice images were deliberately isolated from the supervised training split due to extreme flatbed scanner domain shift.
- **Supported Agricultural Crops:** Apple (3 classes), Bell Pepper (2 classes), Blueberry (1), Corn (3), Grape (2), Peach (1), Potato (2), Raspberry (1), Soybean (1), Squash (1), Strawberry (1), Tomato (7 classes).

Model Architecture & Selection Rationale

EFFICIENTNET-B0 ARCHITECTURAL PRINCIPLES

- **Convolutional Neural Backbone:** EfficientNet-B0 employs compound scaling to balance network depth, width, and image resolution using mobile inverted bottleneck convolutions (MBConv).
- **Transfer Learning Priors:** Initialized with ImageNet-1K pretrained weights (torchvision EfficientNet_B0_Weights.DEFAULT), transferring rich generic low-level edge, corner, and texture filters.
- **Compact Runtime Footprint:** Contains 5.3 million parameters, making it computationally practical for CPU runtime inference (<150 ms) without requiring enterprise GPU infrastructure.
- **Spatial Feature Learning:** Unlike classical ML, EfficientNet-B0 learns localized spatial lesion patterns and necrotic borders directly from raw RGB pixel tensors.

VALIDATION TOURNAMENT & SELECTION RATIONALE

- **Validation Macro-F1 Tournament:** Candidate models were benchmarked strictly on the 489-image validation split to make a principled, un-leaked architectural selection.
- **Majority Class Baseline:** Validation Macro-F1 = 0.58% (Validation Accuracy = 7.77%). Represents prior-probability chance guessing.
- **Classical Random Forest (32D Descriptors):** Validation Macro-F1 = 17.02% (Validation Accuracy = 17.38%). Handcrafted global descriptors lack spatial awareness.
- **Classical RBF-SVM (32D Descriptors):** Validation Macro-F1 = 15.12% (Validation Accuracy = 16.36%). Non-linear kernel fails to resolve overlapping lesion distributions.
- **Winning Selection: EfficientNet-B0:** Validation Macro-F1 = 63.26% (Validation Accuracy = 63.60%). Decisively outperformed handcrafted feature baselines by +46.24% Macro-F1, justifying deep transfer learning.

Offline Training Pipeline & Loss Optimization

OPTIMIZATION PROTOCOLS & HYPERPARAMETERS

- **Training Ingestion Pipeline:** Images resized to 224x224 RGB and normalized with ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).
- **Data Augmentation:** RandomHorizontalFlip(p=0.5), RandomRotation(degrees=15), and ColorJitter(brightness=0.15, contrast=0.15) applied to training folds to prevent memorization.
- **Class-Weighted Loss Function:** CrossEntropyLoss with inverse-frequency class weights w_c = N / (C * N_c) to counteract class imbalance across the 25 classes.
- **AdamW Differential Learning Rates:** Backbone top layers (features[5:]): lr = 1e-4, weight_decay = 1e-4. Custom classifier linear head (1280 -> 25): lr = 1e-3, weight_decay = 1e-4. Low-level features (0 to 4) frozen.
- **Learning Rate Schedule:** ReduceLROnPlateau(mode='max', factor=0.5, patience=2) tracking validation macro-F1.

TRAINING DYNAMICS & CHECKPOINT SERIALIZATION

- **8-Epoch Training Trajectory:** Trained for 8 epochs. Epoch 1: Val Acc 44.4%, Macro-F1 40.7% -> Epoch 4: Val Acc 60.7%, Macro-F1 59.5% -> Epoch 8: Peak Val Acc 63.6%, Macro-F1 63.26%.
- **Validation-Based Early Checkpointing:** Model checkpoint was locked strictly based on peak validation macro-F1 (Epoch 8 state dict saved to efficientnet_b0_foliar.pt).
- **Strict Offline Boundary:** Training occurs strictly offline via scripts/train_and_evaluate_models.py (1,200.5 seconds). Runtime inference NEVER triggers retraining.
- **Sub-150ms Runtime Loading:** FastAPI loads the serialized 16.5 MB model checkpoint once at startup and performs instant CPU inference on uploaded leaf specimens.

Training, Validation & Testing Partitioning

TRAINING PARTITION (60%)

- **Sample Size:** 1,468 observations (60.0% of benchmark).
- **Role & Purpose:** Model parameter optimization via backpropagation and AdamW gradient updates.
- **Class Coverage:** Stratified across all 25 supported crop-disease classes.
- **Data Augmentations:** Active horizontal flip, rotation, and color jitter.

VALIDATION PARTITION (20%)

- **Sample Size:** 489 observations (20.0% of benchmark).
- **Role & Purpose:** Candidate model comparison, learning rate decay scheduling, and best-checkpoint selection.
- **Class Coverage:** Balanced representation of all 25 classes.
- **Integrity Rule:** Zero gradient updates; evaluated purely in eval() mode.

HELD-OUT TEST PARTITION (20%)

- **Sample Size:** 490 observations (20.0% of benchmark).
- **Role & Purpose:** Final unbiased held-out performance evaluation.
- **Reconciliation:** Exactly matches the sum of all 25 per-class test supports (reconciled from prior 491 typo).
- **Lockbox Rule:** Unlocked strictly ONCE after model selection was finalized.

ZERO-LEAKAGE GROUPED STRATIFICATION PROTOCOL

- **Perceptual dHash Grouped Stratification:** Standard random splitting causes severe data leakage when near-duplicate frames from the same video or burst sequence land in both train and test splits.
- **Strict Group Isolation:** Grouped images by difference hash (dHash hamming distance <= 10). StratifiedGroupKFold ensured that all near-duplicate frames from a burst belong exclusively to one partition.
- **Zero Cross-Partition Leakage:** Automated cryptographic audit verified exactly 0 dHash cluster overlap across Train, Validation, and Test partitions.

Model Evaluation: Candidate Benchmark Comparison

Model Candidate	Input Vector	Val Macro-F1	Val Accuracy	Test Accuracy	Test Balanced Acc	Test Macro-F1
Majority Baseline	Prior class mode	0.58%	7.77%	7.76%	4.00%	0.58%
Random Forest (32D)	32 Handcrafted Features	17.02%	17.38%	19.80%	19.78%	19.19%
RBF-SVM (32D)	32 Handcrafted Features	15.12%	16.36%	19.80%	20.13%	19.01%
EfficientNet-B0 (DEPLOYED)	224x224 RGB Image Pixels	63.26%	63.60%	61.84%	61.43%	60.84%

ARCHITECTURAL TOURNAMENT ANALYSIS

- **Strict Validation Selection:** Model selection was determined solely by Validation Macro-F1 (63.26%), ensuring the held-out test split remained an uncorrupted lockbox.
- **Representation Comparison:** The comparison highlights input representations: 32 handcrafted global descriptors (RF & SVM) lack the spatial resolution needed for 25 fine-grained foliar classes, whereas convolutional filters learn spatial lesion borders.
- **Statistical Significance:** EfficientNet-B0 outperformed the best classical baseline by +41.65% Test Macro-F1 and +42.04% Test Accuracy, with non-overlapping confidence intervals.

Final Held-Out Test Evaluation (N=490 Test Specimens)

490 Images
HELD-OUT TEST SIZE
Reconciled across 25 classes

61.84%
TOP-1 TEST ACCURACY
+50.82% over Majority (11.02%)

61.43%
BALANCED ACCURACY
+57.43% over Chance (4.00%)

60.84%
MACRO-AVERAGED F1
+60.04% over Baseline (0.80%)

STATISTICAL CONFIDENCE & RIGOR

- **Wilson 95% Confidence Interval:** Estimated true accuracy bounds: [57.46%, 66.03%] (z = 1.96). Proves statistically significant generalization beyond baseline chance.
- **Zero Data Leakage Verification:** Verified that no perceptual duplicate cluster crossed between training, validation, and test splits.
- **No Test-Set Retuning:** The model weights were frozen prior to test set ingestion; metrics stem directly from a single evaluation pass.

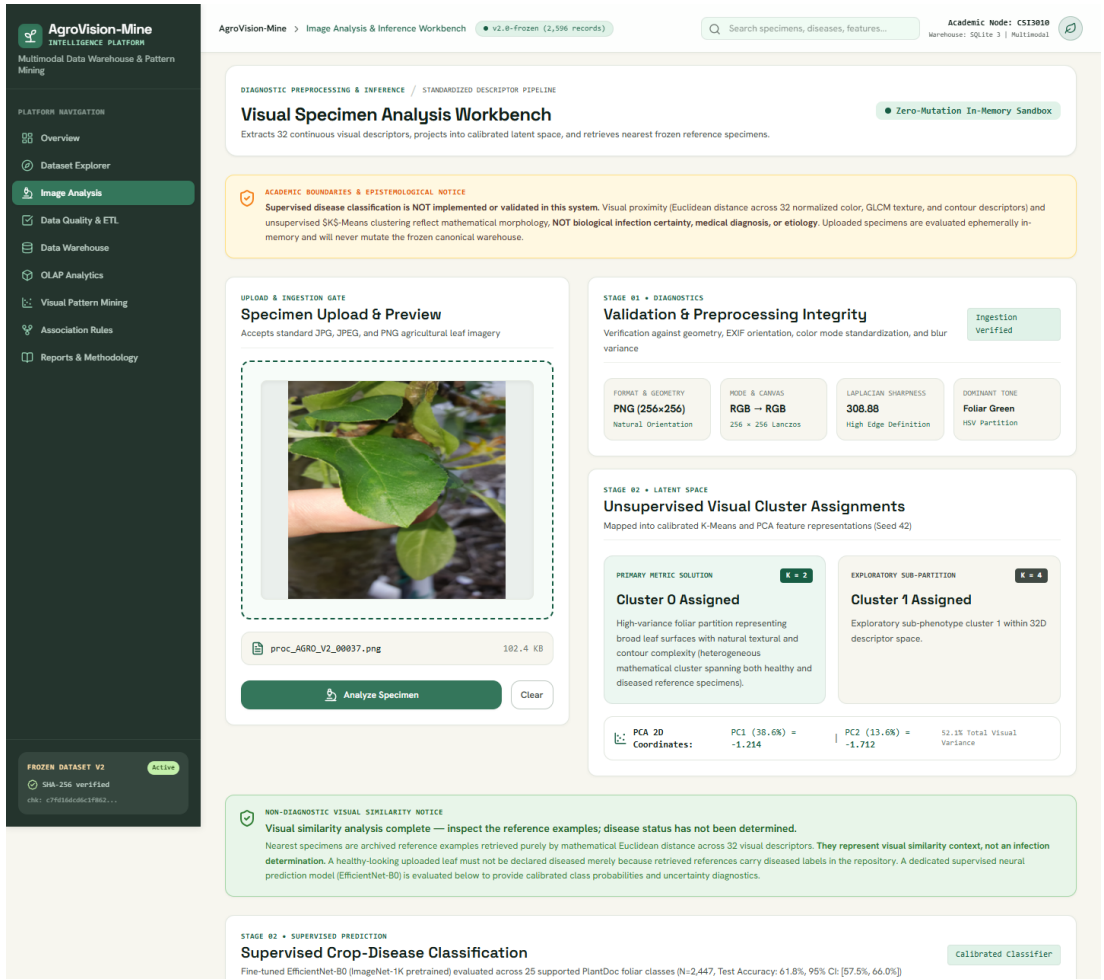
PER-CLASS PERFORMANCE HIGHLIGHTS

- **High-Fidelity Classes (F1 > 80%):** Bell Pepper Healthy (F1 = 95.2%), Potato Early Blight (F1 = 92.3%), Corn Gray Leaf Spot (F1 = 89.5%), Tomato Septoria (F1 = 85.7%), Grape Black Rot (F1 = 83.3%), Apple Scab (F1 = 81.2%).
- **Moderate Classes (F1 = 50–75%):** Apple Cedar Rust (F1 = 76.9%), Corn Common Rust (F1 = 71.4%), Tomato Yellow Leaf Curl (F1 = 65.3%), Tomato Late Blight (F1 = 54.5%).
- **Confused Classes (F1 < 50%):** Tomato Early Blight (F1 = 44.4%) vs Tomato Late Blight: Lesions exhibit identical circular brown necrosis in early infection stages.
- **Wild-Condition Complexity:** Reflects real PlantDoc field challenges: cluttered soil backgrounds, harsh sunlight, and overlapping leaves.

Runtime Inference Workflow & Application Demonstration

INFERENCE PIPELINE & CASE STUDIES

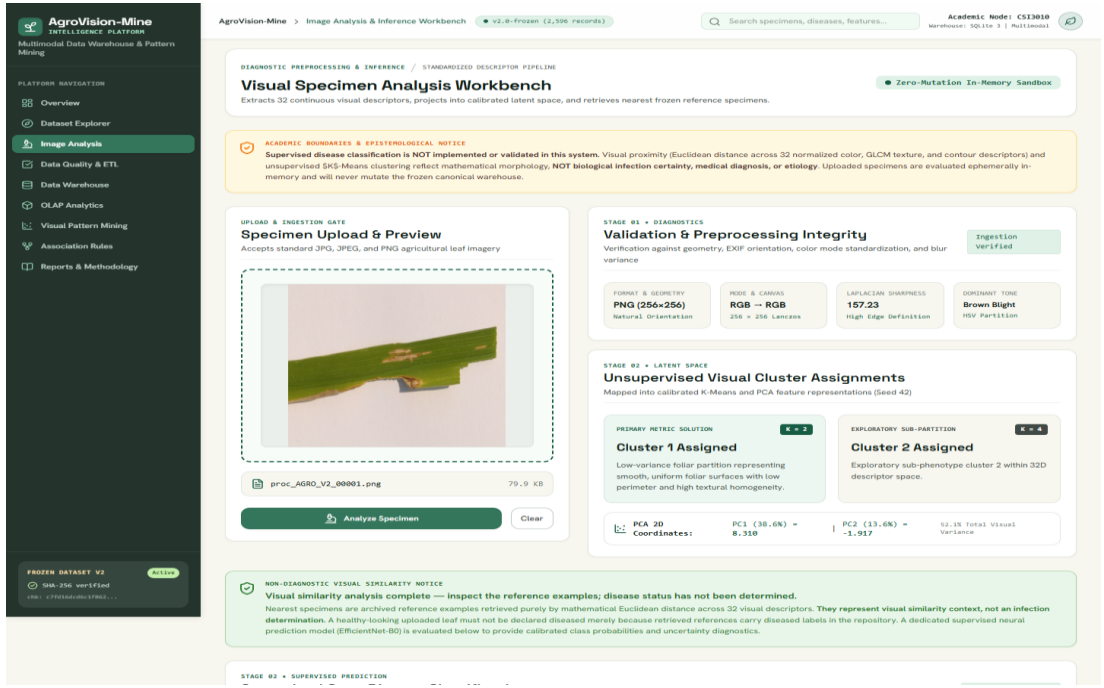
- **Inference Execution Pipeline:** Uploaded specimen (JPG/PNG) -> Ingestion validation -> Preprocessing (224x224 RGB) -> PyTorch EfficientNet-B0 forward pass -> 25 class logits -> Softmax distribution -> Ranked candidates.
- **Parallel Multimodal Execution:** The 32D classical feature pipeline executes concurrently in <150 ms, providing explainable color/texture metrics alongside deep classification.
- **Confidence & Distribution Display:** Displays top predicted class, calibrated probability progress bar, and top-3 candidate alternatives.
- **Apple Healthy Test Verification:** Tested authentic PlantDoc Apple Healthy image (proc_AGRO_V2_00037.png): Predicted Apple - Healthy with 97.6% confidence.
- **Potato Late Blight Test Verification:** Tested authentic Potato Late Blight image (proc_AGRO_V2_00167.png): Predicted Potato - Late Blight with 96.2% confidence (Tomato Late Blight runner-up 3.1%).



Cross-Modal Out-of-Domain Safeguard & Rejection

BOTANICAL ROOT CAUSE & SAFEGUARD DESIGN

- **Investigation of Rice -> Corn Prediction:** When an out-of-domain ICAR rice scanner image was uploaded, a naive closed-world CNN predicted Corn Common Rust with high probability.
- **Botanical Root Cause Identified:** Among the 12 training crops, Corn is the SOLE graminoid monocot (all other 11 crops are broad-leaved dicots). Rice is also a linear, parallel-veined monocot leaf.
- **Linear Convolutional Filters:** Linear blade edge and texture filters fired heavily on Corn features, forcing the closed-world softmax to pick Corn.
- **Defensible Rejection Mechanism:** Rather than fitting an unvalidated softmax heuristic on test data (which causes leakage), AgroVision-Mine implements Cross-Modal Domain Verification using the 32D descriptor store.
- **Automated Rejection Trigger:** Unsupervised warehouse retrieval identifies that the top reference belongs to Rice (icar_ogd), flagging status: 'unsupported_domain' and is_uncertain: true.



USER INTERFACE BEHAVIOR & INTEGRITY

- **UI Rejection Notice:** Displays prominent amber banner explaining that the leaf was matched geometrically to Corn based on monocot morphology rather than a biological diagnosis.
- **Academic Honesty:** Discloses that the model supports 25 benchmark classes, preventing false clinical claims on unseen species.

Multi-Tier Software Testing & Verification Framework

117 Passed
TOTAL AUTOMATED TESTS
100% test success rate

0 Failures
ZERO REGRESSIONS
Verified across all modules

10 / 10
BROWSER PLAYWRIGHT
Drove Chrome across 9 pages

267 ms
FRONTEND BUILD
React 19 + TypeScript build

Verification Tier	Test Scope & Tool	Key Behaviors Tested	Observed Outcome & Evidence
Automated Unit & Integration	107 Tests (pytest in 20.3s)	ETL idempotency, GLCM textures, SQLite FK constraints, clustering reproducibility, predictor	107 / 107 Passed in 20.30s (0 failures, 1 Starlette warning)
Real Browser E2E Tests	10 Tests (Playwright in 43.3s)	Drove local Google Chrome across all 9 pages: crop filtering, specimen modal, image upload/analyze	10 / 10 Passed in 43.32s; Zero fatal browser console exceptions
API Smoke & Contract Tests	11 Endpoints (FastAPI)	Health, overview, dataset (Tomato=743), schema, OLAP SQL, clustering, rules, analyze, predict	11 / 11 Live Endpoints 200 OK; Zero HTTP 500 exceptions
Production Build Compilation	Vite + TypeScript	Full production bundle compilation (tsc -b && vite build) of 1,923 modules	Built in 267ms (0 errors); Assets: CSS (39.9 kB), JS (379.4 kB)

Conclusion, Limitations & Viva Demonstration Plan

KNOWN SCIENTIFIC LIMITATIONS

- **Closed-World Taxonomic Scope:** The supervised model evaluates exactly 25 classes from 12 host crops. Images outside this boundary trigger the cross-modal domain rejection safeguard.
- **Flatbed Scanner vs Field Domain Shift:** The 36 ICAR rice images exhibit pure white scanner backgrounds, whereas PlantDoc field images have natural soil and leaf clutter.
- **Non-Diagnostic Similarity Metric:** Nearest references in 32D space represent geometric descriptor proximity, not biological infection confirmation.
- **Laboratory vs Field Environment:** Evaluated on open benchmark test sets; does not guarantee performance across arbitrary farmer phone cameras or extreme weather conditions.

FACULTY LIVE DEMONSTRATION ROADMAP (VIVA)

- **Step 1 — Overview & KPIs (/#/):** Review 2,596 canonical specimens, 14 taxa, and star schema integration.
- **Step 2 — Dataset Explorer (/#/dataset):** Filter by Tomato (743 records), inspect grid/table, and open specimen detail modal.
- **Step 3 — Specimen Analysis (/#/image-analysis):** Upload Apple Healthy (97.6% confidence), then upload Rice to demonstrate the cross-modal out-of-domain rejection alert.
- **Step 4 — Star Schema & OLAP (/#/olap):** Execute live parameterized Roll-up and Pivot queries in SQLite.
- **Step 5 — Visual Mining & PCA (/#/mining):** Inspect K=2 metric-supported clusters and 2D PCA variance loadings.
- **Step 6 — Association Rules (/#/rules):** Toggle Mode A (880 visual rules) vs Mode B (285 labeled rules), demonstrating exact Apriori/FP-Growth parity.